

LOON LAKE, SK, Canada

WMO: 712730

Lat: 54.0197N

Lon: 109.1383W

Elev: 546

StdP: 94.94

Time Zone: -6.00 (W06)

Period: 09-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-33.5	-30.3	-37.1	0.1	-33.2	-33.8	0.2	-30.0	7.4	-8.2	6.5	-9.9	1.1	280	0.626

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.6	27.2	17.0	25.6	16.4	24.1	15.8	18.8	24.8	17.8	23.6	16.8	22.1	3.5	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.5	12.5	21.5	15.5	11.8	20.5	14.6	11.1	19.4	55.6	24.7	52.4	23.6	49.4	22.1	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.3	6.3	5.6	DB	-37.8	30.0	3.0	1.1	-40.0	30.8	-41.7	31.5	-43.4	32.1	-45.6	33.0	
(5)			WB	-37.9	21.3	3.0	1.4	-40.1	22.3	-41.8	23.1	-43.5	23.9	-45.7	24.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	2.0	-12.9	-13.4	-5.6	2.5	10.4	14.6	17.0	15.5	10.8	3.3	-6.0	-13.1		
	DBStd	12.83	9.07	8.38	8.08	5.48	4.57	3.10	2.45	2.88	4.48	4.81	7.02	8.51		
	HDD10.0	3584	709	654	483	230	52	4	0	2	44	212	479	716		
	HDD18.3	5978	967	887	742	476	247	115	56	93	227	466	729	974		
	CDD10.0	675	0	0	0	5	65	143	217	172	68	4	0	0		
	CDD18.3	28	0	0	0	0	2	4	14	6	1	0	0	0		
	CDH23.3	484	0	0	0	2	64	76	173	112	57	0	0	0		
	CDH26.7	66	0	0	0	0	8	9	25	12	12	0	0	0		
(14) Wind	WSAvg	2.5	2.5	2.5	2.8	3.1	2.8	2.6	2.3	2.1	2.4	2.6	2.6	2.4		
(15) Precipitation	PrecAvg	441	16	12	21	22	42	83	74	66	48	14	16	19		
	PrecMax	599	43	35	60	65	162	243	187	163	111	41	47	51		
	PrecMin	321	2	1	2	0	3	26	18	6	11	1	0	6		
	PrecStd	76	9	9	15	16	32	49	35	35	25	10	13	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	6.9	13.1	22.4	27.8	27.9	29.3	28.2	28.7	20.1	11.6	6.1		
		MCWB	3.2	2.4	5.8	10.1	13.8	16.4	19.7	18.3	17.1	11.2	6.3	2.3		
	2%	DB	4.9	3.8	9.6	17.7	25.3	25.5	26.9	26.0	24.7	15.9	7.9	3.7		
		MCWB	1.8	0.7	4.0	7.7	12.8	15.8	18.6	17.8	15.3	8.9	3.7	0.7		
	5%	DB	3.0	1.5	6.8	14.7	22.5	23.6	25.2	24.5	21.7	13.3	5.0	1.3		
		MCWB	0.5	-0.9	2.4	6.5	11.8	14.9	17.2	16.8	13.9	7.4	2.2	-1.1		
	10%	DB	0.7	-0.8	4.6	11.6	20.4	21.9	23.7	22.9	19.1	10.9	2.9	-1.3		
		MCWB	-1.2	-2.3	1.2	4.9	10.9	13.9	16.5	15.8	12.7	6.6	0.8	-2.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.4	3.2	6.2	10.2	15.4	18.4	21.1	20.1	17.6	12.1	6.8	2.7		
		MCDB	7.0	6.0	12.3	21.8	22.2	24.6	27.7	26.1	26.2	18.5	10.8	5.5		
	2%	WB	2.0	0.8	4.3	8.2	13.9	17.1	19.1	18.6	16.0	9.8	4.0	0.8		
		MCDB	4.8	3.6	8.9	16.6	22.0	23.1	25.2	24.3	23.7	13.5	7.3	3.6		
	5%	WB	0.7	-0.6	2.7	6.9	12.7	16.0	18.1	17.5	14.5	8.2	2.3	-1.1		
		MCDB	2.9	1.4	6.2	13.8	21.2	21.9	24.1	22.9	20.4	12.6	4.6	1.1		
	10%	WB	-1.2	-2.4	1.4	5.4	11.5	15.1	17.2	16.5	13.1	6.7	0.8	-2.9		
		MCDB	0.8	-0.7	4.3	10.9	19.2	20.2	22.2	21.5	18.4	10.7	2.6	-1.2		
(35) Mean Daily Temperature Range	MDBR		10.3	11.4	10.8	10.9	13.8	11.5	11.6	11.9	12.0	9.3	8.0	9.1		
	5% DB	MCDBR	10.7	11.8	12.8	16.7	18.1	14.8	14.6	15.2	17.4	14.1	11.0	10.9		
		MCWBR	8.3	9.0	8.1	8.9	8.1	6.9	7.3	8.1	8.9	8.2	7.1	8.5		
	5% WB	MCDBR	10.7	11.4	11.5	15.6	16.1	12.3	13.6	13.9	16.6	13.3	9.9	10.5		
		MCWBR	8.5	9.0	7.4	8.6	7.5	6.2	7.3	7.9	8.9	8.0	6.9	8.4		
(40) Clear-Sky Solar Irradiance	taub		0.238	0.258	0.294	0.320	0.344	0.363	0.365	0.361	0.309	0.281	0.249	0.213		
	taud		2.332	2.306	2.334	2.421	2.398	2.361	2.382	2.373	2.519	2.568	2.466	2.356		
	Ebn at Noon		758	854	892	913	903	885	875	857	872	831	750	731		
	Edh at Noon		53	77	95	101	109	115	111	105	80	61	47	42		
(44) All-Sky Solar Radiation	RadAvg		0.90	1.95	3.47	4.68	5.71	5.75	5.90	4.81	3.37	1.93	0.87	0.63		
	RadStd		0.09	0.13	0.24	0.39	0.49	0.42	0.38	0.39	0.38	0.32	0.12	0.06		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page